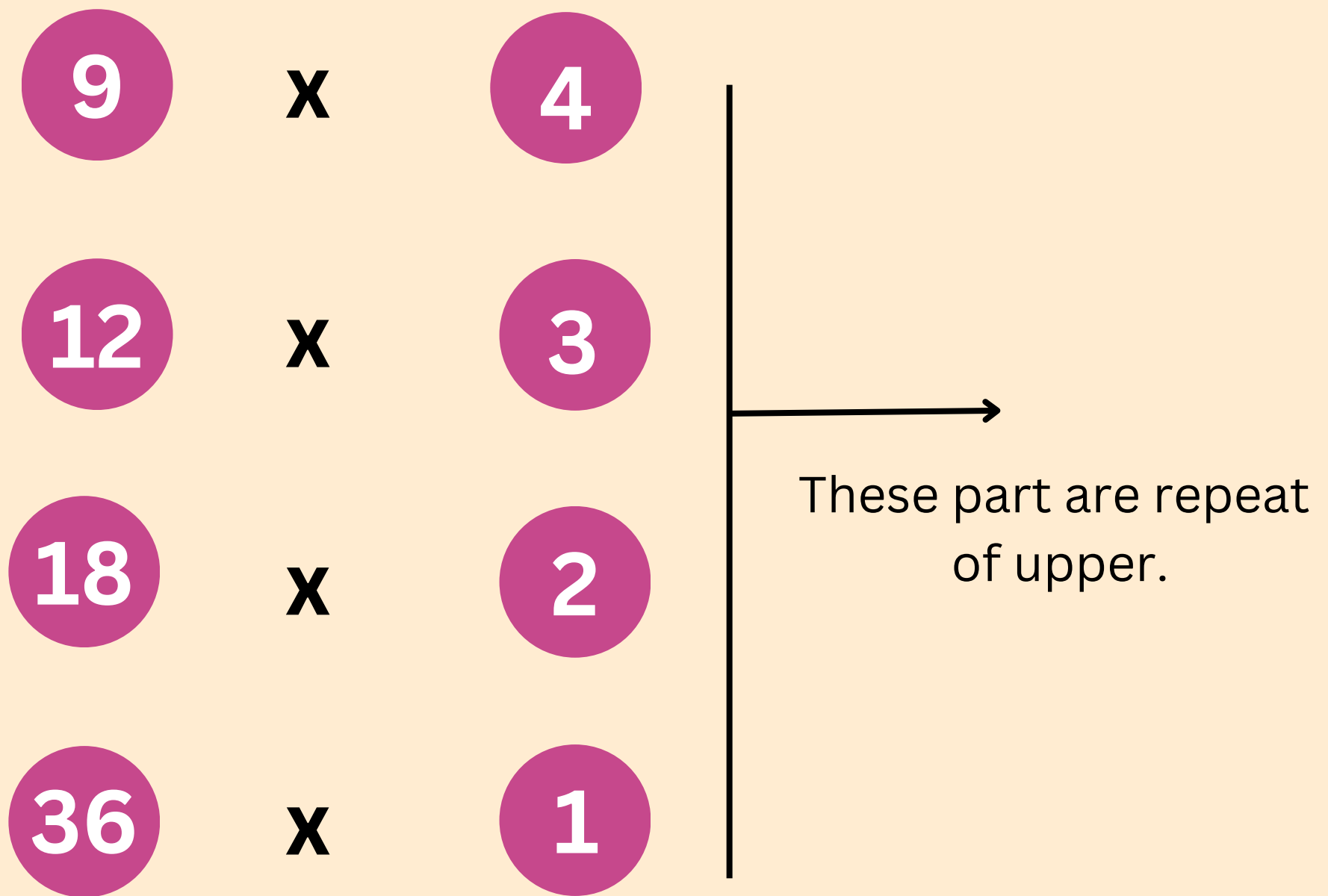


Mathematical observation of divisor problem

$$N = 36$$

1	x	36
2	x	18
3	x	12
4	x	9
6	x	6

These part are repeat
of lower.



- **$\text{sqrt}(N)$ = repeat part**

e.g: $\text{sqrt}(36) = 6$

code:

```
9  #include<bits/stdc++.h>
10 using namespace std;
11
12 void printAllDivisors(int N) {
13     if (N <= 0)
14         return;
15
16     for (int i = 1; i <= sqrt(N); i++) {
17         if (N % i == 0) {
18             cout << i << " ";
19             if (N / i != i) {
20                 cout << N / i << " ";
21             }
22         }
23     }
24 }
25
26 int main() {
27     int N = 36;
28     printAllDivisors(N);
29     cout << endl;
30
31     return 0;
32 }
33
```



1 36 2 18 3 12 4 9 6

- find no of divisors:

```
#include <iostream>
#include <cmath>

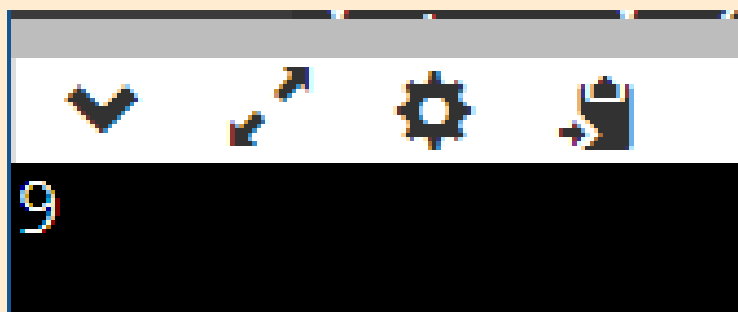
using namespace std;

int numberOfDivisors(int N) {
    if (N <= 0)
        return 0;

    int count = 0;
    for (int i = 1; i <= sqrt(N); i++) {
        if (N % i == 0) {
            count++;
            if (N / i != i) {
                count++;
            }
        }
    }
    return count;
}

int main() {
    int N = 36;
    cout << numberOfDivisors(N) << endl;

    return 0;
}
```





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